

A CERTIFIED FINITE CENSUS FOR ADDITIVE-SQUARE-FREE WORDS ON FOUR-LETTER INTEGER ALPHABETS OF HEIGHT AT MOST FIVE

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ABSTRACT. An additive square is a pair of consecutive blocks having the same positive length and the same sum. The existence of an infinite additive-square-free word over a finite integer alphabet remains open. We give a certified finite census of every primitive normalized four-letter integer alphabet $\{0 < a < b < c\}$ with $c \leq 5$, modulo reflection. In particular, we obtain

$$g(\{0, 1, 2, 5\}) = 86, \quad g(\{0, 1, 3, 5\}) = 88,$$

where $g(A)$ is the maximum length of an additive-square-free word over A . Both values, the other five census entries, and the extremal-word counts are reconstructed by a second implementation that imports no discovery output. We also certify that a fixed point of a 2-uniform endomorphism on four symbols cannot solve the infinite problem under any integer weighting. These are finite and restricted-class results; they do not decide the infinite existence question.

1. INTRODUCTION

For a finite set $A \subset \mathbb{Z}$, call a finite or infinite word over A *additive-square-free* (ASF) if it has no factor UV with $|U| = |V| > 0$ and $\sum U = \sum V$. The question whether some finite integer alphabet supports an infinite ASF word arose in the language of uniformly repetitive semigroups [5]. Recent primary sources continue to state the finite-integer question as open [6, 1, 9].

Several neighboring theorems are important but have different quantifiers. Brown proved unavoidable bounded-error approximations to additive squares [2]. Rao and Rosenfeld constructed an infinite ASF morphic word with weights in $\mathbb{Z}^2$, rather than $\mathbb{Z}$ [7]. Freedman and Brown determined exact finite maxima for four-element sets satisfying $a + d = b + c$ and proved that the minimum of $g(A)$ over all four-element real sets is 50 [3]. A public exact computation later gave $g(\{0, 1, 2, 4\}) = 62$ [8]; we reproduced that computation before extending the census.

Our main result is deliberately finite. After affine normalization and reflection there are seven four-letter integer alphabets of height at most five. We close all seven ASF prefix trees using exact integer arithmetic and reconstruct them using an independent evaluator. The two unbalanced

height-five values, 86 and 88, were not located in our dated result-specific literature search. This wording is not a priority claim.

2. DEFINITIONS AND REDUCTIONS

Definition 2.1. For a nonempty finite alphabet $A \subset \mathbb{Z}$, put

$$g(A) = \sup\{|w| : w \in A^* \text{ is ASF}\} \in \mathbb{N} \cup \{\infty\}.$$

The empty word and all one-letter words are ASF; the two compared blocks must have positive length.

Lemma 2.2 (Affine normalization). *If $\alpha, \beta \in \mathbb{Z}$ and $\alpha \neq 0$, applying $x \mapsto \alpha x + \beta$ to each letter preserves and reflects the ASF property.*

Proof. For equal-length blocks U, V ,

$$\sum(\alpha U + \beta) - \sum(\alpha V + \beta) = \alpha(\sum U - \sum V);$$

the translation terms cancel and α is nonzero. $\square$

Thus an integer alphabet with at least two elements has a primitive representative

$$A = \{0 = x_0 < x_1 < \cdots < x_{k-1}\}, \quad \gcd(x_1, \dots, x_{k-1}) = 1.$$

The reflection $x \mapsto x_{k-1} - x$ gives a further involution. Arbitrary relabeling is not allowed because sums depend on the integer labels.

If $S_0 = 0$ and $S_n = \sum_{j < n} a_j$, then a factor starting at i with half-length ℓ is an additive square exactly when

$$(2.1) \quad S_i + S_{i+2\ell} = 2S_{i+\ell}.$$

For one fixed finite A , König's infinity lemma applied to the ASF prefix tree shows that $g(A) = \infty$ exactly when an infinite ASF word over A exists. It is essential that A remain fixed.

There cannot be a bound on $g(A)$ depending only on $|A|$. Keränen constructed an infinite abelian-square-free word on four symbols [4]. Taking a prefix of length $N - 1$ and assigning the four symbols the values $1, N, N^2, N^3$ gives an ASF word: if two block sums were equal, the difference of their Parikh vectors would give $\sum_{j=0}^3 d_j N^j = 0$ with $|d_j| < N$. Reducing after division by the lowest nonzero power of N gives a contradiction modulo N . Hence $g(\{1, N, N^2, N^3\}) \geq N - 1$, as also observed by Freedman and Brown [3].

3. THE CERTIFIED CENSUS

Theorem 3.1 (Four-letter census through height five). *Let A be a four-element integer alphabet whose primitive normalized representative has maximum at most 5. Up to reflection, its exact value $g(A)$ and the complete-tree*

statistics are as follows.

A	$g(A)$	nodes	leaves	maximizers
$\{0, 1, 2, 3\}$	50	187,789	50,136	16
$\{0, 1, 2, 4\}$	62	19,097,778	5,350,440	2
$\{0, 1, 3, 4\}$	55	659,727	183,150	4
$\{0, 1, 2, 5\}$	86	299,596,819	83,646,960	8
$\{0, 1, 3, 5\}$	88	234,485,149	66,026,525	2
$\{0, 1, 4, 5\}$	55	1,137,005	322,450	4
$\{0, 2, 3, 5\}$	55	850,735	241,212	4

The empty root is included in the node count.

Proof. For $A = \{0 < a < b < c\}$ with $c \leq 5$, direct enumeration subject to $\gcd(a, b, c) = 1$ and replacement by the lexicographically smaller of A and $c - A$ gives one class at $c = 3$, two at $c = 4$, and four at $c = 5$; these are exactly the seven rows displayed.

For each row, traverse the rooted tree whose vertices are ASF words and whose children are one-letter ASF extensions. Suppose the parent word is ASF. Appending a letter leaves every old factor unchanged, so a new forbidden factor, if one exists, must end at the new final position. Testing all possible positive suffix half-lengths using exact prefix sums therefore accepts exactly the ASF children. The reported traversal closes every branch. Consequently its maximum depth is an upper bound for $g(A)$.

For the matching lower bound, the traversal records all words at maximum depth. A separate program, using right-tail cumulative sums rather than the discovery program's prefix sums, reconstructs the full tree from the serialized alphabet, regenerates the extremal words, and obtains the same node count, leaf count, maximum depth, extremal count, and digest of the ordered extremal list. This exact replay proves each row. $\square$

For inspection, one maximizer in each new height-five class is

$\{0, 1, 2, 5\}$: 02050125051012101501025152505251202515015201
215120105010201521201525150201025202515202,
 $\{0, 1, 3, 5\}$: 10503050103101503515030105153505351353015135
10503051013010503050151315053513101351505315.

Their respective lengths are 86 and 88. The serialized certificate contains one witness and a digest binding the complete extremal list for every row.

4. A RESTRICTED MORPHIC NO-GO THEOREM

Proposition 4.1. *Let h be a 2-uniform endomorphism of a four-symbol alphabet, prolongable on a letter. Its fixed point contains an abelian square ending within its first 19 symbols. Hence no integer weighting of such a four-state fixed point is ASF.*

Proof. Relabel the prolongable letter as 0. Then $h(0) = 0x$, leaving seven unrestricted symbol positions among the four length-two images. There are therefore exactly $4^7 = 16,384$ morphisms. For each one, generate a 128-symbol fixed-point prefix and scan endpoints in increasing order, comparing the Parikh vectors of every pair of adjacent equal-length blocks. Every morphism has a witness; the latest first endpoint is 19, attained by six morphisms. A second implementation enumerates the seven positions again and uses direct multiset comparisons, reproducing the endpoint histogram and the sixfold extremal count. Finally, every abelian square has equal sums under every integer weighting. $\square$

The proposition concerns four-state endomorphisms only. It does not cover a coding of a uniform fixed point with more hidden states, a nonuniform morphism, or uniform modulus at least three.

5. CERTIFICATE AND REPRODUCIBILITY

The principal certificate is `certificates/census_max5.json`. Its SHA-256 digest is

`ebeb72ca4f04e377de266f4c403f6e2480bed3232feabb923d1436685057ce19.`

Run

```
python3 certificates/verify_finite_census.py \
  certificates/census_max5.json
```

to compile the independent C++20 evaluator in a temporary directory and reconstruct all seven records. The expected success record includes

```
"records": 7, "status": "VERIFIED"
```

and the independent evaluator digest

`dc8c5e25393fd58456ab72178b79cd636ec6f4006ccdfd80a4b363e1c766ef19.`

The schema rejects missing or unexpected fields, noninteger numeric claims, nonprimitive alphabets, inconsistent witness counts, and malformed hashes. Seven corruption tests are included in `tests/test_fail_closed.py`, including deletion of an entire canonical alphabet record.

For Proposition 4.1, run

```
python3 certificates/verify_uniform2_no_go.py \
  certificates/uniform2_no_go.json
```

The computations use signed exact integers and no randomness. The discovery and certification sources share definitions but not executable state or outputs.

6. DIAGNOSTICS AND LIMITATIONS

As a positive-construction diagnostic, we projected the Rao–Rosenfeld $\mathbb{Z}^2$ fixed point onto scalar integer directions. In its first 30,000 symbols, all 12,176 primitive normalized directions of coefficient height at most 100

acquire a scalar additive square. This is a bounded exact screen, not a theorem about directions of larger height.

The main open problem is unchanged. Theorem 3.1 covers neither normalized height at least six nor alphabets with at least five letters. Proposition 4.1 covers only one morphism class. A finite prefix surviving a search would not prove infinite avoidance without an all-length invariant or a valid finite-ancestor theorem.

DATA, CODE, AND COMPUTATIONAL DISCLOSURE

The accompanying source package contains the formal statement, literature ledger, route and gap ledgers, discovery programs, serialized certificates, independent verifiers, negative tests, build logs, and a SHA-256 release manifest. OpenAI Codex was used for literature search, mathematical exploration, code and verifier drafting, exact computation, audit, and manuscript drafting. All theorem-level computational claims were rerun by a separate no-import implementation. No proof assistant was used.

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